LECTURE 1

Knowledge, Representation and Reasoning

Contents

Knowledge and its types

Data, information and knowledge

Knowledge representation

Reasoning and its types

Knowledge-based systems and practice

Objectives

Define knowledge, representation and reasoning.

Explain knowledge types and their link to intelligence.

Distinguish data, information and knowledge.

Explain why AI systems need representation.

Apply a rule to derive a conclusion.

Identify these components in an AI scenario.

Opening Questions: Knowledge and Intelligence

1

What makes a computer system intelligent?

2

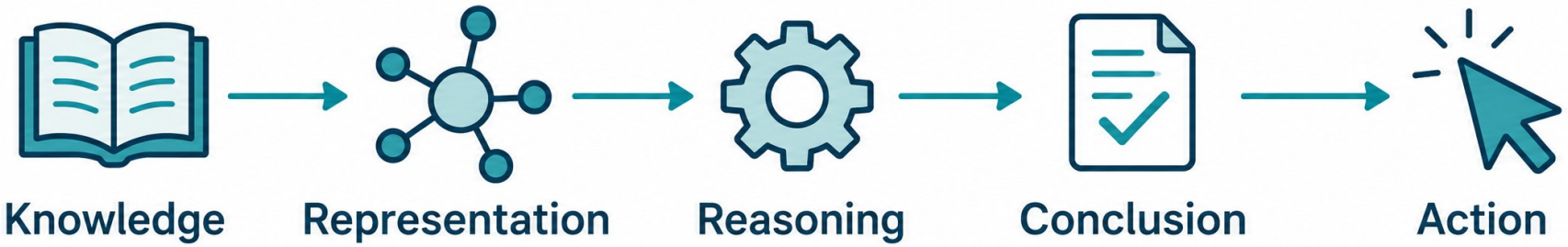
Does storing a large amount of information automatically make a computer intelligent?

3

What do we mean by knowledge?

Intelligent Behavior in AI

An AI system uses what it knows to answer questions, solve problems, or choose actions.



What does it know?
How is that knowledge expressed?
What follows from it?

Knowledge in AI

Knowledge is information about a domain that a system can use to answer questions, solve problems, or make decisions.

Fact

Ali is a student.

Student(Ali)

Rule

Every student has an ID.

Student(x) → HasID(x)

Knowledge can also describe relationships, categories, properties, and constraints.

Types of Knowledge

Type	Meaning	Example
Factual	Facts or situations	Ali is a student.
Conceptual	Concepts and categories	A student is a type of person.
Relational	Connections between entities	Studies(Ali, ComputerScience)
Rule-based	Conditions and conclusions	Passing all required courses permits graduation.

Each type helps the system describe or reason about its domain.

Data, Information and Knowledge

Concept	Meaning	Example
Data	Raw values or observations	25
Information	Data with meaningful context	Ali is 25 years old.
Knowledge	Facts and rules used to understand or infer	Given the rule “age > 18 means adult,” infer that Ali is an adult.

Context gives data meaning. Applying a rule supports a conclusion.

Traffic Data and Route Decisions

Data

A sensor counts 95 vehicles.

Information

Road A has heavy traffic.

Knowledge

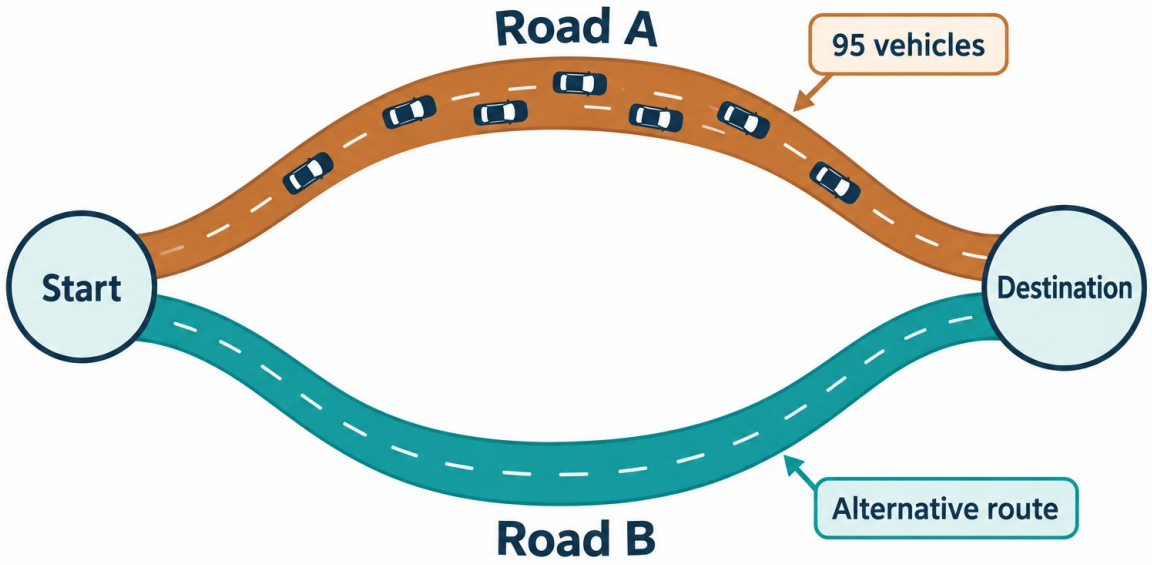
Heavy traffic increases travel time.

Reasoning

Road B may offer a faster route.

Decision

Recommend Road B if evidence supports it.



Classroom example. Vehicle icons are symbolic.

Traffic System: Knowledge in Use

A sensor reports 95 vehicles on Road A. The system interprets heavy traffic, uses a travel-time rule, and recommends an alternative route.

Discussion

At which stage does the system begin to use knowledge?

Explain your choice using the traffic example.

Knowledge Representation

Knowledge representation expresses knowledge in a structured or formal form that a computer can use for reasoning.

Natural language

All students who pass the examination receive a certificate.

Formal rule

$\text{Passes}(x) \rightarrow \text{ReceivesCertificate}(x)$

Specific fact

$\text{Passes}(\text{Ali})$

x stands for an object. The arrow means “if ... then ...”.

Components of a Representation

Component	Role	Example
Entities	Things in the domain	Ali, Sara, Car1
Properties	Characteristics of entities	Red(Car1)
Relations	Connections between entities	Studies(Ali, ComputerScience)
Rules	Conditions for conclusions	Student(x) → HasID(x)
Categories	Membership and generalization	Student(x) → Person(x)

Logic and Production Rules

Logic

Represents facts and relationships using logical statements.

Human(x) → Mortal(x)

If something is human, it is mortal.

Production rules

Expresses what to do when a condition holds.

IF temperature > 30°C
THEN turn on cooling

Both approaches make a condition and its consequence explicit.

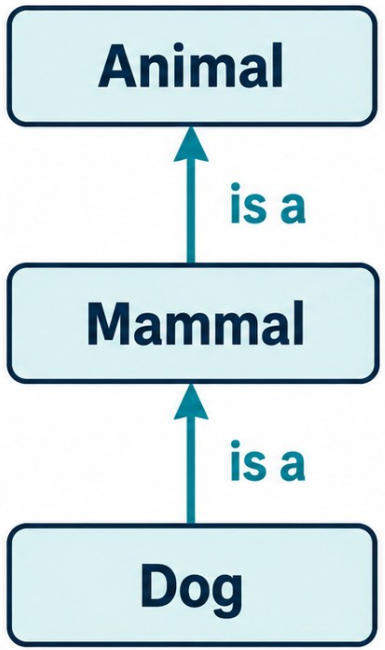
Frames, Inheritance and Ontologies

FRAME

Student
Name
Age
Department
Courses
RegistrationNo

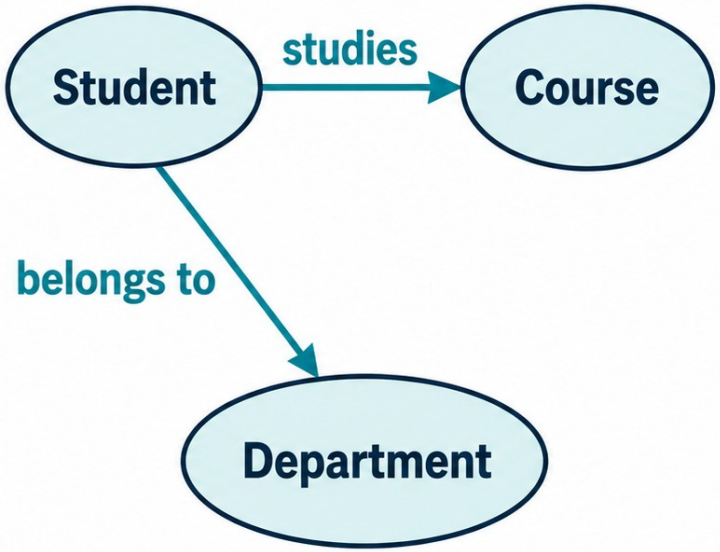
Attributes of an object

INHERITANCE



Category relationships

ONTOLOGY



Domain concepts
and their relationships

Representation: Storage and Questions

1

Why can’t we simply store everything as raw data?

2 Given this knowledge

Studies(Ali, ComputerScience)

What does Ali study?

Why Representation Matters

Organization

Connect facts about students, courses, and universities.

Reasoning

Use a student fact and an ID rule to derive a new conclusion.

Question answering

Use Studies(Ali, ComputerScience) to answer what Ali studies.

Problem solving

Use domain relationships to investigate possible causes.

Decision making

Use conditions and rules to recommend an action.

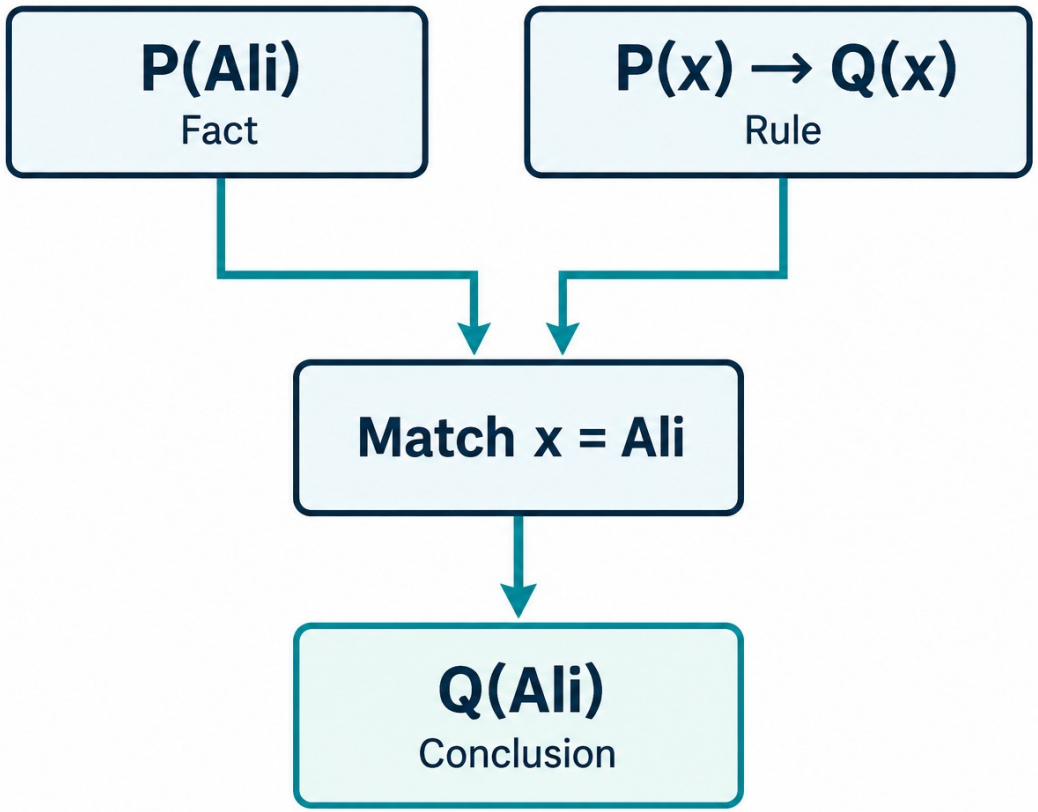
Reasoning with Known Facts

Reasoning

Uses available knowledge to determine what follows.

Rule application

Match the fact to the condition, then substitute Ali for x.



The matching fact and the rule support the conclusion.

Deductive Practice: Two Conclusions

Case 1: Socrates

Human(x) → Mortal(x)
Human(Socrates)

What can we conclude?

Case 2: Ahmed

CSStudent(x)
→ StudiesProgramming(x)
CSStudent(Ahmed)

What can we conclude?

For each case, identify the rule and the matching fact.

Deduction and Derived Knowledge

Deduction applies a general rule to a specific fact.

Explicit knowledge

The system receives the rule and fact.

Human(x) → Mortal(x)
Human(Socrates)

Derived knowledge

The system infers the conclusion.

Mortal(Socrates)

Socrates is mortal.

A valid deductive step preserves truth when the premises are true.

Worked Example: Ahmed Studies Programming

1 Identify the rule

$CSStudent(x) \rightarrow StudiesProgramming(x)$

2 Find the matching fact

$CSStudent(Ahmed)$

3 Substitute the individual

$x = Ahmed$

4 Derive the conclusion

$StudiesProgramming(Ahmed)$

Ahmed studies programming.

Practice: Reasoning Through Categories

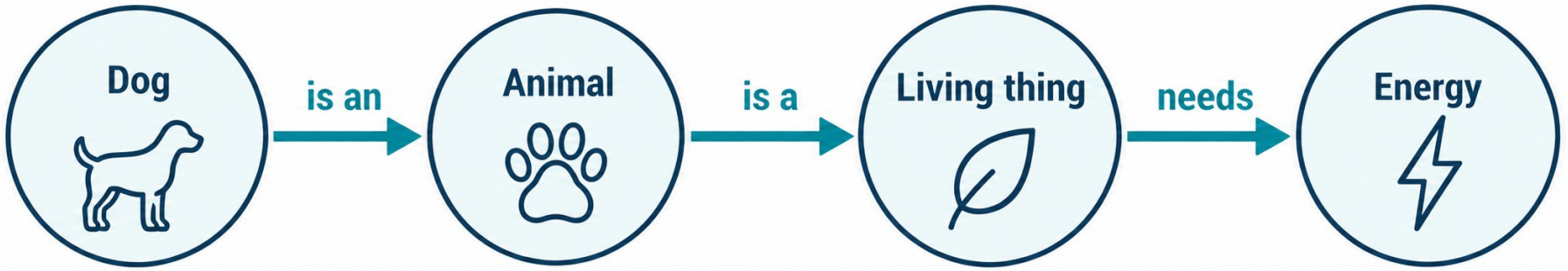
Given these relationships:

Dog(x) → Animal(x)
Animal(x) → LivingThing(x)
LivingThing(x) → RequiresEnergy(x)

Does a dog require energy?

Show the reasoning chain that supports your answer.

Reasoning Across Several Rules



Dog(x) → Animal(x)
Animal(x) → LivingThing(x)
LivingThing(x) → RequiresEnergy(x)

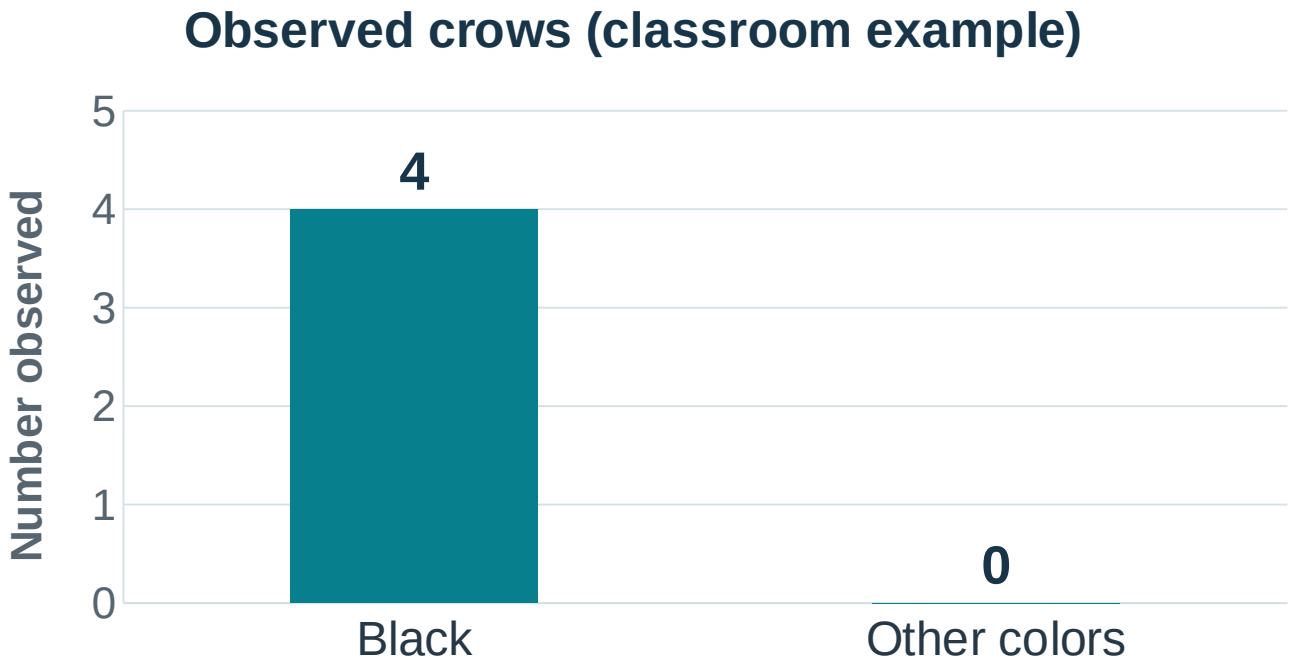
Derived rule

Dog(x) → RequiresEnergy(x)

A conclusion from one rule can support the next reasoning step.

Inductive Reasoning

Induction uses observations to support a generalization.



Observation

All four observed crows are black.

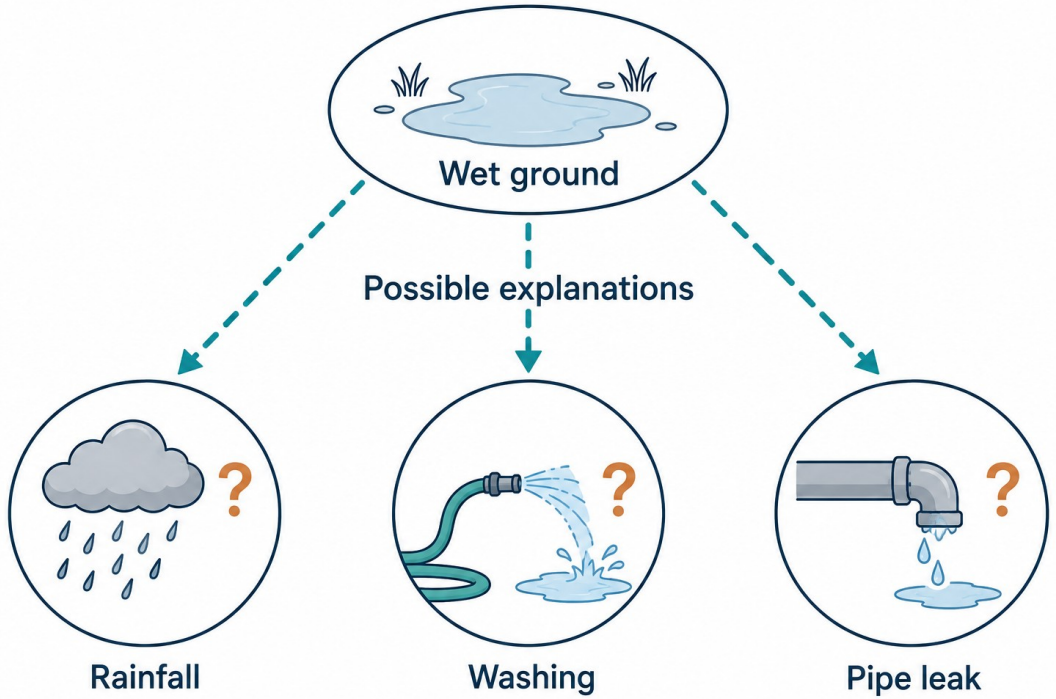
Generalization

Crows are generally black.

Four observations do not prove that every crow is black.

Abductive Reasoning

Abduction seeks a plausible explanation for an observation.



A weather report indicating rain can support the rainfall explanation.

Comparison of Reasoning Types

Type	Reasoning pattern	Main question	Status of conclusion
Deductive	Rule and fact to conclusion	What must follow?	Entailed if the premises hold and inference is valid
Inductive	Observations to generalization	What generalization is supported?	Supported, but not guaranteed
Abductive	Observation to explanation	What could explain this?	Plausible, but not guaranteed

The reasoning method determines what kind of claim the conclusion supports.

Discussion: Reasoning Without Knowledge

Suppose a system has a powerful reasoning mechanism but no domain knowledge.

Can it provide useful medical advice?

Justify your answer by identifying what the system is missing.

Knowledge and Reasoning Work Together

Knowledge without reasoning

Stored rules may leave useful conclusions undiscovered.

Reasoning without domain knowledge

A mechanism lacks the facts and rules needed for a useful answer.

Their combined role

Represent relevant knowledge, derive a conclusion, and choose an action.

Advising Practice: Course Registration

Student(Ali)
Completed(Ali, ProgrammingFundamentals)
Requires(DataStructures, ProgrammingFundamentals)

Rule

If a student has completed all prerequisites, the student can register for the course.

Can Ali register for Data Structures?

Assume this is the only prerequisite and no other restriction applies.

University Advising Example

Ali has completed the prerequisite for Data Structures.

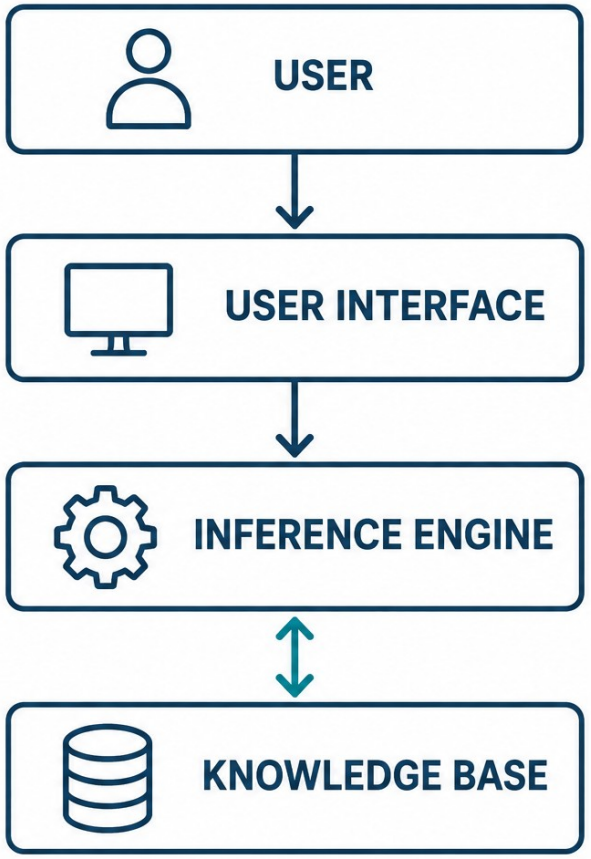


Student(Ali)
Completed(Ali, ProgrammingFundamentals)
Requires(DataStructures, ProgrammingFundamentals)

Recommendation: Ali can register under the example's rule.

Assume this is the only prerequisite and no other registration restriction applies.

Knowledge-Based System Architecture



Knowledge base

Stores represented facts, rules, and relationships.

Inference engine

Applies reasoning to derive conclusions.

User interface

Receives questions and presents answers.

Troubleshooting with a Knowledge Base

Known fact

BatteryLow(Car1)

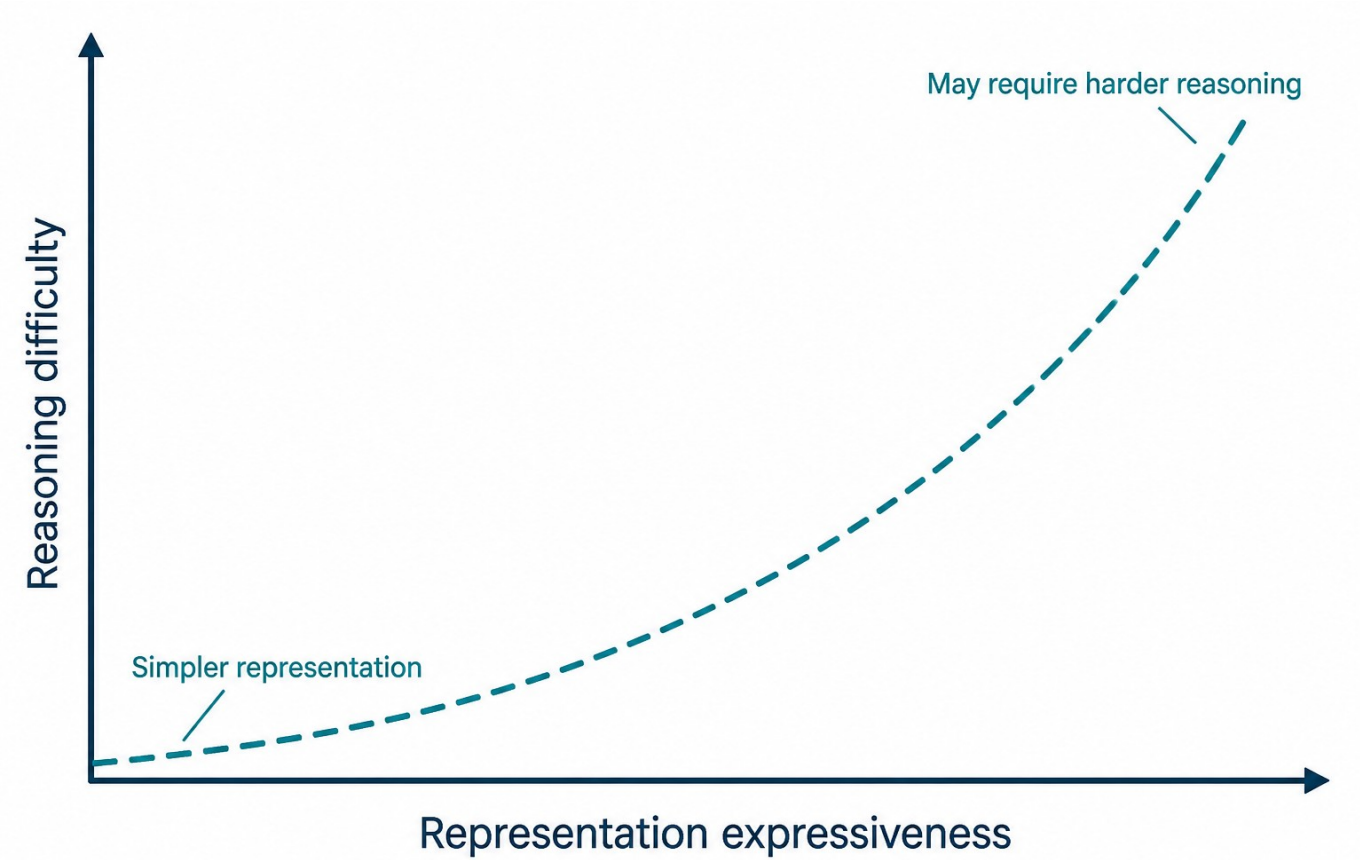
Rules

BatteryLow(x) → EngineMayNotStart(x)
EngineMayNotStart(x) → CheckBattery(x)

Derived conclusions

EngineMayNotStart(Car1)
CheckBattery(Car1)

Expressiveness and Reasoning Difficulty



Greater expressiveness

Can describe more complex situations, exceptions, and uncertainty.

Simpler representations

Can make reasoning easier, while leaving some details unexpressed.

Conceptual illustration. Difficulty depends on the representation and reasoning task.

Classroom Activity: Exam-Hall Entry

Facts

Registered(Sara) **ValidID(Sara)**

Rules

Registered(x) → HasUniversityID(x)

HasUniversityID(x) ∧ ValidID(x)
→ CanEnterExamHall(x)

∧ means AND: both conditions must hold.

What can the system conclude about Sara?
Which conclusions require more than one condition?

Exam-Hall Activity: Discussion Questions

1

Which statements were given directly?

2

Which statement was inferred?

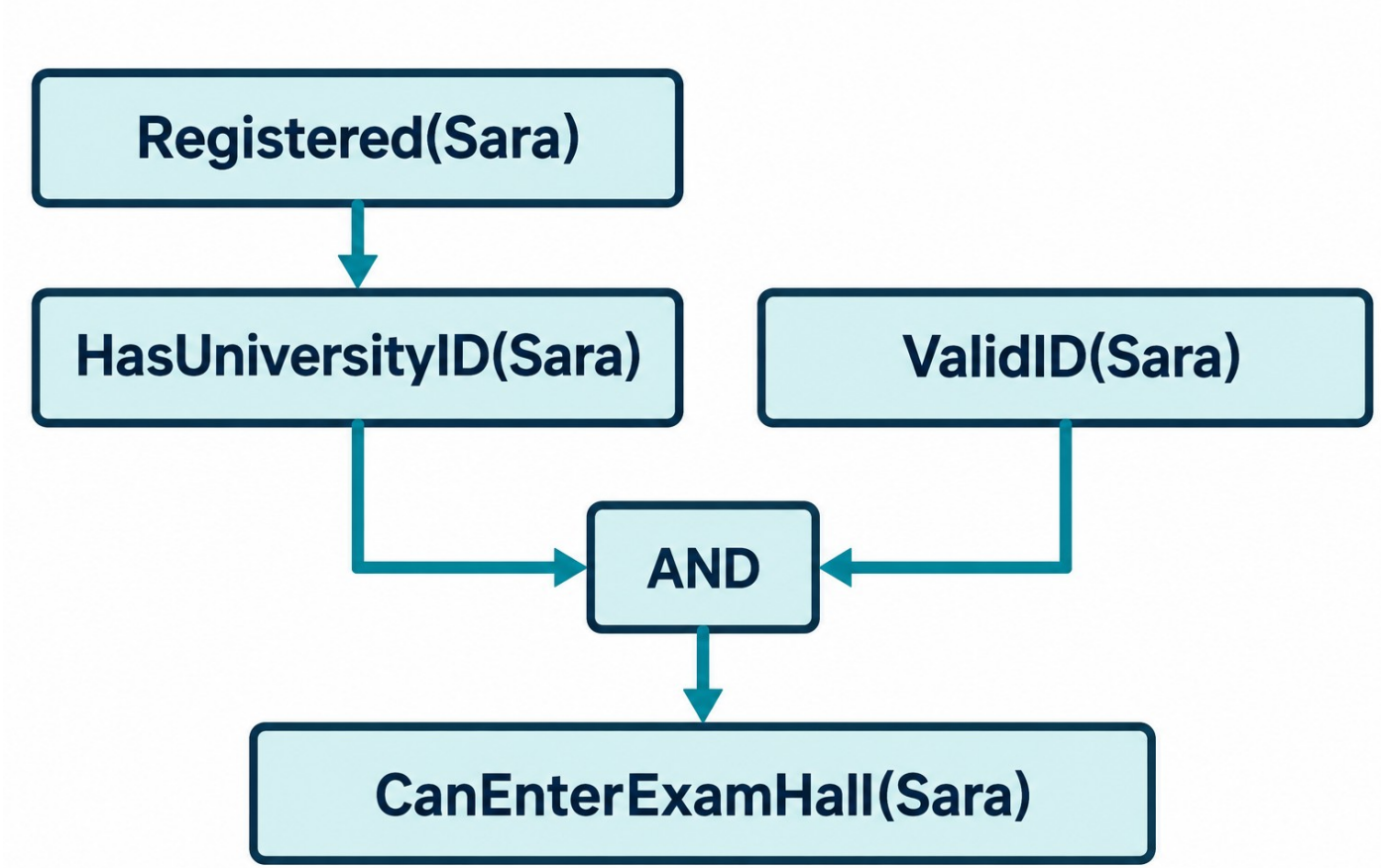
3

Which rule was used?

4

What would happen if Sara’s ID were not valid?

Exam-Hall Entry: Reasoning and Conditions



Two conditions

Sara must have a university ID, and that ID must be valid.

Unmet condition

This rule cannot establish permission if validity is unmet.

A derived fact can combine with an explicit fact in the next rule.

Common Misconceptions

Misconception	Clarification
Representation is just storing data.	Structure lets a system use and reason with knowledge.
Reasoning means calculation.	Reasoning also draws conclusions and explanations.
AI is only machine learning.	AI also includes representation and reasoning.
Every conclusion is certain.	Certainty depends on the premises and inference method.
Facts and conclusions have the same origin.	The system receives explicit facts and infers conclusions.

Bloom Questions: Concepts and Rule Application

Remember

What is knowledge representation?

Understand

Explain why knowledge representation is important for AI.

Apply

Student(Ali)
Student(x) → HasID(x)

Derive the conclusion.

Bloom Questions: Analysis and Evaluation

Analyze

Consider an AI traffic system. Identify which parts represent data, knowledge, reasoning, and decision-making.

Evaluate

Why might simply storing more data not necessarily make an AI system more intelligent? Justify your answer.

How should knowledge be represented so that useful reasoning is possible?

Scenario Analysis: Academic Advising

A university AI advising system knows that Ahmed is a BSCS student, that Data Structures requires Programming Fundamentals, and that Ahmed has completed Programming Fundamentals. The system recommends that Ahmed register for Data Structures.

1. Identify the knowledge.
2. Identify how the knowledge can be represented.
3. Identify the reasoning performed.
4. Explain how the conclusion supports intelligent behavior.

Assume this is the only prerequisite and no other registration restriction applies.

Formative Check: Knowledge Concepts

Question 1

Define knowledge in the context of AI.

Question 2

Differentiate between data, information, and knowledge.

Question 3

Why is knowledge representation necessary?

Formative Check: Inference and Reasoning Types

Question 4: Given

Human(x) → Mortal(x)
Human(Ali)

What can be concluded?

Question 5

Explain the difference between deductive, inductive, and abductive reasoning.

Formative Check: Smart Traffic

Recall the classroom traffic example:

Sensors report 95 vehicles on Road A. The system interprets heavy traffic. It knows that heavy traffic generally increases travel time, considers Road B, and recommends an alternative route.

Question 6

Analyze a smart traffic system and identify where knowledge and reasoning are used.

Key Takeaways

Knowledge

What a system knows about its domain.

Representation

The structured form in which it expresses knowledge.

Reasoning

The process that derives conclusions from knowledge.

Intelligent behavior

Useful answers or actions supported by those conclusions.

Knowledge-based system

A knowledge base working with an inference mechanism.

Homework: Concepts and Comparison

Part A: Explain in your own words

Knowledge
Knowledge representation
Reasoning
Knowledge-based system

Part B: Prepare a comparison table

Compare data, information, and knowledge.

Include a definition and one example of each.

Homework: Reasoning About Tweety

Part C: Given

Bird(Tweety)
Bird(x) → Animal(x)
Animal(x) → LivingThing(x)

Determine what can be concluded about Tweety.

Show your reasoning step by step.

Homework: Analysis of an AI Application

Part D: Select one AI application

Medical diagnosis, autonomous vehicle, university advising, smart traffic management, or robotics.

1. What knowledge does the system require?
2. How could that knowledge be represented?
3. What reasoning could the system perform?
4. What intelligent action could result?

Exit Ticket

1 Complete the relationship

Knowledge + _____ supports intelligent behavior.

2 State the conclusion

Student(Ali) **Student(x) → HasID(x)**

3 Explain in one sentence

Why do we need knowledge representation?